

Austin A. MBaye

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Research Interests: Topological Data Analysis · Computational Biology · Time Series Analysis · Computer Vision

EDUCATION

Ph.D. in Mathematics · Northeastern University
Boston, MA

Aug 2024 – Present

- Advisor: [Dr. Jose Perea](#)
- Lakshmibai Venkatramani Fellowship

B.A. in Mathematics (Minor: Education) · Vassar College
Poughkeepsie, NY

Aug 2020 – May 2024

- Advisor: [Dr. Adam Lowrance](#)
- Departmental Honors in Mathematics · Sigma Xi · GPA: 3.82/4.0

RESEARCH EXPERIENCE

Visiting Researcher – Max Planck Institute of Molecular Cell Biology and Genetics (MPI-CBG)
Dresden, Germany

Jun 2026 – Jul 2026

- Under Dr. Heather Harrington, and Dr. Stephan Grill, applied **Chiral Multiparameter Persistence (CaMPers)** to analyze actomyosin cortical flow fields and quail nematic fields, quantifying symmetry breaking and directional bias in embryonic cell dynamics.

Graduate Research Assistant – Perea Lab, Northeastern University
Boston, MA

Sep 2024 – Present

- Developed **Chiral Multiparameter Persistence (CaMPers)**, a novel TDA framework for detecting and quantifying chirality in vector fields on Riemannian manifolds; proved stability theorems and introduced chiral Euler Characteristic Curves (cECC).
- Developed **AQSM_SW1PerS**, a persistent homology pipeline for automated quantification of stereotypical motor movements in autism from multimodal sensor data; validated on real-world behavioral datasets.

Undergraduate Researcher – MSRI Undergraduate Program (MSRI-UP)
Berkeley, CA

Jun 2023 – Jul 2023

- Selected as one of 18 students nationwide for a competitive NSF-funded REU at the Mathematical Sciences Research Institute.
- Co-designed a persistent cohomology algorithm using cup products to detect quasi-periodicity in univariate time series; implemented the full pipeline in Python.
- Presented findings at MSRI and co-authored a technical report distributed to program participants.

PUBLICATIONS AND PREPRINTS

Publications

1. **MBaye, A.A.**, Perea, J.A., Tralie, C.J., & Goodwin, M.S. *Automated quantification of stereotypical motor movements in autism using persistent homology*. *Sci Rep*, 2026. <https://doi.org/10.1038/s41598-026-60095-8>
Software: **AQSM_SW1PerS** Dataset: [figshare doi:10.6084/m9.figshare.30100669.v1](https://figshare.com/dataset/figshare/30100669.v1)

RESEARCH PRESENTATIONS

Invited Talks

- *Automated Quantification of Stereotypical Motor Movements in Autism*. 5th International Conference on Neuroscience & Psychiatry, Paris, France. Nov. 2025. [\[link\]](#)
- *Automated Quantification of Stereotypical Motor Movements in Autism*. Mathematics Colloquium, Vassar College, Poughkeepsie, NY. Jan. 2024. [\[link\]](#)

Contributed Talks

- *Chiral Multiparameter Persistence*. Joint Mathematics Meetings, Chicago, IL. Jan. 2027. (Scheduled)

- *Automated Quantification of Stereotypical Motor Movements in Autism*. Joint Mathematics Meetings, Chicago, IL. Jan. 2027. (Scheduled)
- *Chiral Multiparameter Persistence*. AMS Fall Western Sectional Meeting, Tempe, AZ. Nov. 2026. (Scheduled)
- *Automated Quantification of Stereotypical Motor Movements in Autism*. AMS Spring Eastern Sectional Meeting, Hartford, CT. Apr. 2025. [\[link\]](#)
- *Using Persistent Cup Products for Quasi-Periodicity Detection*. Joint Mathematics Meetings, San Francisco, CA. Jan. 2024. [\[link\]](#)
- *Using Persistent Cup Products for Quasi-Periodicity Detection*. MSRI-UP Final Presentation, Berkeley, CA. Jul. 2023. [\[link\]](#)

Posters

- *Using Persistent Cup Products for Quasi-Periodicity Detection*. SACNAS National Diversity in STEM Conference, Portland, OR. Oct. 2023.

SOFTWARE

- CaMPers – Python/C++ package for computing chiral Euler characteristic curves from flow fields. (In development)
- [AQSM_SW1PerS](#) – Python package implementing a persistent homology pipeline for automated quantification of stereotypical motor movements from behavioral video and accelerometer data.

AWARDS AND FELLOWSHIPS

- **Lakshmibai Venkatramani Fellowship** for Mathematics Graduate Students – Northeastern University (2026)
- **MSRI-UP Fellowship** – Mathematical Sciences Research Institute (2023)
- **Sigma Xi** – Scientific Research Honor Society, inducted Vassar College chapter (2024)
- **Departmental Honors in Mathematics** – Vassar College (2024)